

9.1.1.5. Are loc proprietatea de absorbție?

$$p \uparrow (p \downarrow q) \equiv p$$

	P	q	$p \downarrow q$	$p \uparrow (p \downarrow q)$
i ₁	T	T	F	T
i ₂	T	F	F	T
i ₃	F	T	F	T
i ₄	F	F	T	T

Nu are loc prop. de absorbție deoarece

$$i_3: \{p, q\} \rightarrow \{T, F\}$$

$$i_3(p) = F$$

$$i_3(q) = F$$

$$i_3(p \uparrow (p \downarrow q)) = T \neq F \equiv i_3(p)$$

Cele 2 forme nu sunt echivalente

9.1.4.5. semidistributivitatea la stânga a conectivei „ $\vee$ ” față de „ $\rightarrow$ ”

$$p \vee (q \rightarrow r) \rightarrow ((p \vee q) \rightarrow (p \vee r))$$

	P	q	r	$q \rightarrow r$	$p \vee q$	$p \vee r$	$p \vee (q \rightarrow r)$	$(p \vee q) \rightarrow (p \vee r)$	$((p \vee q) \rightarrow (p \vee r))$
i ₁	T	T	T	T	T	T	T	T	T
i ₂	T	T	F	F	T	T	T	T	T
i ₃	T	F	T	T	T	T	T	T	T
i ₄	T	F	F	T	T	T	T	T	T
i ₅	F	T	T	T	T	T	T	T	T
i ₆	F	T	F	F	T	F	F	F	T
i ₇	F	F	T	T	F	T	T	T	T
i ₈	F	F	F	T	F	F	T	T	T

9.1.3.5.

$$p \rightarrow q \models (q \rightarrow r) \rightarrow (p \rightarrow q \vee r)$$

	p	q	r	$p \rightarrow q$	$q \rightarrow r$	$q \vee r$	$p \rightarrow q \vee r$	$(q \rightarrow r) \rightarrow (p \rightarrow q \vee r)$
i_1	T	T	T	T	T	T	T	T
i_2	T	T	F	T	F	T	T	T
i_3	T	F	T	F	/	/	/	/
i_4	T	F	F	F	/	/	/	/
i_5	F	T	T	T	T	T	T	T
i_6	F	T	F	T	F	T	T	T
i_7	F	F	T	T	T	T	T	T
i_8	F	F	F	T	T	F	T	T

Se obs. că $\forall i \in \{i_1, i_2, i_5, i_6, i_7, i_8\}$, unde $i(u) = T \Rightarrow$
 $\Rightarrow i(v) = T$, deci $u \models v$